

Wafer-Rack Equal- Q Navier–Stokes Study

Audit, Correction, and Reproducibility Check

CFD code review & reproduction

October 2, 2026

Abstract

A two-dimensional incompressible Navier–Stokes solver for flow through a rack of six wafers is reviewed, corrected, and re-run. The solver uses a geometry-aligned Marker-and-Cell (MAC) staggered grid with a fractional-step (projection) method, first-order donor-cell upwind advection and explicit second-order diffusion. An equal-rack-flow constraint fixes the inlet velocity of each pitch by a bracketed scalar root solve. Reproduction reruns *through the same solver implementation* recover the audited production inlet speeds and hydraulic resistances, confirm discrete mass conservation at roundoff ($\sim 10^{-13}\%$), and recover the ordering $R_{h,\text{Small}} > R_{h,\text{Medium}} > R_{h,\text{Large}}$. This is a reproducibility check, not an independent-implementation verification. A preliminary grid/timestep study is reported for the Small case only (§6): timestep error is negligible and grid refinement is monotonic, but the continuum-limit error is not formally estimated, and Medium/Large remain unrefined. The absolute U_{in} , Δp and R_h values are therefore discretization-dependent (see §10).

Nomenclature

Symbol	Description	Units (2-D)
<i>Geometry</i>		
W	Tank width	m
H	Tank height	m
N_w	Number of wafers (= 6)	–
P	Wafer pitch (centre-to-centre spacing)	m
t_h	Wafer thickness (= 0.015 00 m)	m
y_0, y_1	Lower / upper wafer extent (0.12, 0.25)	m
w_{in}	Inlet slot width (= 0.015 00 m)	m
y_{cut}	Horizontal flux-evaluation line (= 0.20)	m
<i>Field variables</i>		
u, v	Horizontal / vertical velocity components	m s^{-1}
$\mathbf{u}$	Velocity vector (u, v)	m s^{-1}
$ \mathbf{u} $	Flow speed $\sqrt{u^2 + v^2}$	m s^{-1}
p	Kinematic pressure (p/ρ)	$\text{m}^2 \text{s}^{-2}$
U_{in}	Prescribed inlet (slot) vertical speed	m s^{-1}
<i>Fluid / control parameters</i>		
ν	Kinematic viscosity (= 7.500 00 $\times 10^{-4} \text{ m}^2 \text{s}^{-1}$)	$\text{m}^2 \text{s}^{-1}$

Symbol	Description	Units (2-D)
Q_{rack}	Flow rate through the five interior gaps at y_{cut}	$\text{m}^2 \text{s}^{-1}$
Q_0	Equal-flow target ($= 0.0055463$)	$\text{m}^2 \text{s}^{-1}$
Q_{in}	Total inlet flux at the bottom boundary	$\text{m}^2 \text{s}^{-1}$
$Q(y)$	Net flux across horizontal face y	$\text{m}^2 \text{s}^{-1}$
Δp	Rack pressure drop ($p_{\text{below}} - p_{\text{above}}$)	$\text{m}^2 \text{s}^{-2}$
R_h	Hydraulic resistance $\Delta p / Q_{\text{rack}}$ (kinematic Δp)	s^{-1}
<i>Discretization</i>		
N_x, N_y	Number of pressure cells in x, y	—
n_{gap}	Fluid cells across each inter-wafer gap ($= 20$)	—
$\Delta x, \Delta y$	Local cell sizes (nonuniform)	m
Δt	Explicit time step	s
res	Steady-state residual (rel. velocity change)	—
<i>Operators / acronyms</i>		
∇, ∇	Discrete divergence / gradient operators	m^{-1}
MAC	Marker-and-Cell staggered grid	—
FD	(legacy) Finite-difference model	—
CFL	Courant–Friedrichs–Lewy number	—

1 Governing equations and method

The flow solves the 2-D incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

on a geometry-aligned MAC staggered grid. A fractional-step projection advances the velocity: an explicit predictor (first-order donor-cell upwind advection, second-order nonuniform diffusion) is followed by a pressure Poisson solve and a divergence-free correction,

$$\nabla^2 p = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^*, \quad \mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla p. \quad (2)$$

The discrete gradient (correction) and divergence operators are mutually adjoint, so the assembled pressure matrix is exactly $-\nabla \cdot \nabla$ of the operators actually applied; this guarantees discrete mass conservation. The pressure Poisson system is factorized once per grid and reused across all root-finding evaluations. Boundary conditions: no-slip on the tank side/bottom walls and all wafer surfaces, prescribed vertical inlet slots at the bottom, and a $p = 0$ open outlet at the top.

Equal- Q control. For each pitch the inlet speed U_{in} is chosen so the rack flow equals a common target Q_0 . This is a bracketed secant iteration with a bisection safeguard, where every function evaluation is a CFD solve advanced to an approximate steady state (see §3 for the acceptance criterion and its limitations). It enforces an integral control constraint; it is not a fit to data.

2 Code audit and corrections

The audited bundle was reviewed statically and then exercised. Corrections fall in two groups.

Numerical corrections (change the computed results).

1. **Inlet grid alignment.** Originally the inlet slots were selected by testing whether a bottom cell *centre* lay within $w_{\text{in}}/2$ of a slot centre; because the 0.015 m edges were not grid faces, the realized discrete slot widths differed by pitch (0.0150/0.0135/0.0130 m for Small/Medium/Large, i.e. up to 13% undersized). The gap construction now places both slot edges as exact x -grid faces, so every discrete slot is 0.0150 m and the total active inlet width is 0.0750 m for all pitches.
2. **Steady-state gate tightened and enforced.** The acceptance residual was tightened ($3 \times 10^{-5} \rightarrow 10^{-5}$) and the observable-drift tolerance tightened ($10^{-3} \rightarrow 2 \times 10^{-4}$, now including Δp); reaching the step cap raises an error instead of silently returning. Equal- Q evaluations are run best-effort while rooting and convergence-enforced on the final polish, with an enforced secant/bisection correction, with a final tolerance assertion, so the accepted steady-state Q is within tolerance. The same correction loop and assertion are used in both the production driver and the verification harness.

Non-numerical corrections (do not change the discretization).

1. **Hard-coded /mnt/data paths removed** (driver and the verification/plot scripts); output goes to \$WAFER_OUT (default: the script directory), created lazily.
2. **Self-test fixed.** Solver(0.06, 0.0025) $\rightarrow$ Solver(0.06, 20) (n_{gap} is an integer cell count).

3. **Driver de-duplication.** `run_production_equalq.py` is now a thin re-export; single-case runners share one `metrics`.
4. **Stale documentation relabelled** (legacy FD speeds 0.5605/0.3000/0.1833) and the near-wall v y -diffusion asymmetry documented in place.
5. **Checksum workflow.** `gen_checksums.sh` regenerates `SHA256SUMS.txt`; the manifest is regenerated once the numerical version is final.

3 Steady-state acceptance and its limitations

A run is accepted as “converged” when, after a minimum number of steps, the per-step relative velocity change falls below a residual threshold *and* the chunk-to-chunk relative change of the acceptance observables (Q_{rack} , central peak v , Δp) falls below a drift tolerance; reaching the step cap raises an error rather than silently returning. The acceptance thresholds govern how tightly each equal- Q evaluation is settled: too loose a gate can stop a cold-started solve short of the root tolerance (a large-pitch cold start was observed to stop at -0.14% before further iteration reached -0.004%), although the warm-started production chain ultimately lands within tolerance. The criterion is therefore stated as “advanced to an approximate steady state,” and the gate here is tightened relative to the original bundle.

4 Pressure-tap definition

The rack pressure drop is *not* measured immediately across the wafer surfaces. In `metrics`, $\Delta p = p_{\text{below}} - p_{\text{above}}$, where each p is the gap-width-weighted average of rack-fluid pressure on the single cell row nearest a requested plane ($y = 0.10$ below, $y = 0.28$ above). The realized plane positions are grid-dependent (on the Large grid they are $y \approx 0.0975$ and 0.2825). Consequently the absolute $R_h = \Delta p / Q_{\text{rack}}$ is pressure-tap-definition dependent; moving the planes by one or two cells shifts R_h by several percent. The resistance *ordering* across pitches is robust to this choice, but the absolute values should be read together with the tap definition stated here.

5 Reproduction results

Table 2 lists reruns through the audited driver’s own convergence and root-finding logic (the *same* solver implementation; this is a reproducibility check, not an independent-implementation verification). These are reproduction runs (each within the 0.075% root tolerance), reported separately from the final production values. Discrete mass conservation holds to roundoff and the resistance ordering is recovered.

Table 2: Reproduction reruns (corrected code, same solver). $Q_0 = 0.0055463$.

Case	P	U_{in}	Q_{rack}	$(Q_{\text{rack}} - Q_0) / Q_0$	Δp	R_h
Small	0.040	0.69381	$5.546\,90 \times 10^{-3}$	+0.011%	0.081760	14.740
Medium	0.060	0.35199	$5.546\,80 \times 10^{-3}$	+0.010%	0.015134	2.7283
Large	0.080	0.20643	$5.547\,20 \times 10^{-3}$	+0.016%	0.005473	0.9867

Corrected (grid-aligned 0.0750 m inlet), enforced convergence. Δp taps at $y \approx 0.0975/0.2825$ (§4). Inlet alignment lowered U_{in} for Medium (-9.4%) and Large (-13.0%) while Small was already slot-aligned; R_h (a rack-only ratio) is essentially unchanged.

Table 3: Discrete mass conservation: worst section-flow error $\max_y |Q(y) - Q_{\text{in}}|/Q_{\text{in}}$.

Case	max section-flow error
Small	$8.900\,00 \times 10^{-14} \%$
Medium	$7.800\,00 \times 10^{-14} \%$
Large	$6.700\,00 \times 10^{-14} \%$

All three cases are within the 0.075% root tolerance under enforced convergence, and mass conservation holds to roundoff. The hydraulic resistances (14.740/2.728/0.987) are essentially unchanged from the pre-alignment baseline (14.739/2.730/0.987), confirming that R_h is insensitive to the inlet-width correction; the inlet speeds, by contrast, drop for Medium and Large once the slots are identically 0.0750 m wide.

6 Grid and timestep convergence (Small case)

The Small case (narrowest pitch, highest speeds, largest R_h) is the stress case. The equal- Q solution was re-rooted on three meshes ($n_{\text{gap}} = 12, 16, 20$ cells across the smallest clear gap) and the timestep was halved at fixed U_{in} on the production mesh.

Table 4: Small-pitch grid refinement (equal- Q re-rooted per mesh).

n_{gap}	$N_x \times N_y$	Δx_{min}	Δy_{min}	U_{in}	R_h	central peak v
12	186×63	$1.670\,00 \times 10^{-3}$	$4.170\,00 \times 10^{-3}$	0.69542	15.014	0.08219
16	218×85	$1.500\,00 \times 10^{-3}$	$3.080\,00 \times 10^{-3}$	0.69298	14.835	0.08142
20	238×106	$1.250\,00 \times 10^{-3}$	$2.500\,00 \times 10^{-3}$	0.69382	14.740	0.08130

The sequence is monotonic; refinement from $n_{\text{gap}} = 16$ to 20 changes R_h by -0.64% (peak v by -0.15%). This is the last successive-grid difference, *not* an estimate of the remaining continuum-limit error: with only three nonuniform, not strictly self-similar grids and a scheme containing first-order terms, the continuum error is not formally quantified here (a crude $h \propto 1/n_{\text{gap}}$ extrapolation would place the finest R_h roughly 1.5–2.6% above an extrapolated limit). A further systematically refined mesh would be needed to assign a defensible numerical uncertainty.

Table 5: Small-pitch timestep halving at fixed U_{in} ($n_{\text{gap}} = 20$).

	Q_{rack}	central peak v	Δp
change when $\Delta t \rightarrow \Delta t/2$	+0.0001%	+0.003%	−0.017%

The explicit scheme is effectively timestep-independent at the nominal Δt (diffusive limit, advective CFL < 1).

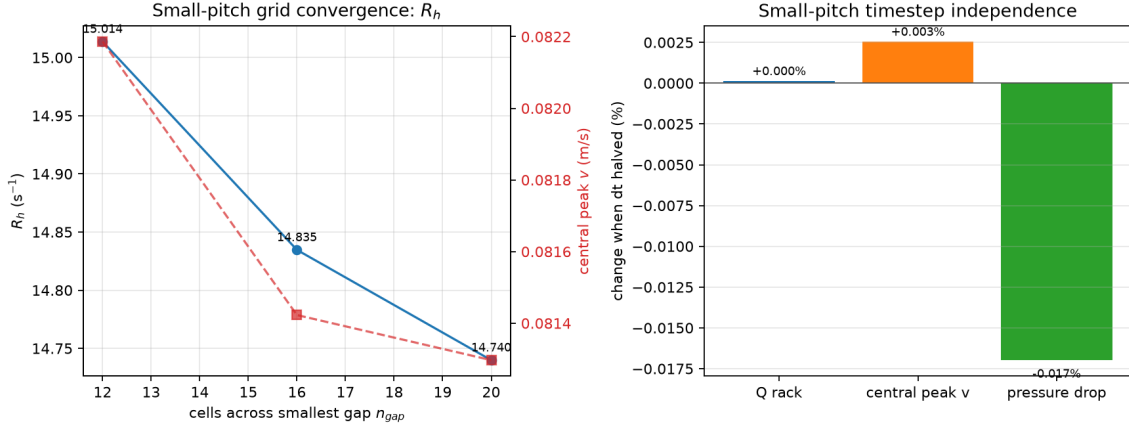


Figure 1: Left: R_h (blue) and central peak v (red) versus mesh resolution, converging with refinement. Right: relative change of observables when the timestep is halved – all below 0.02%.

7 Flow fields

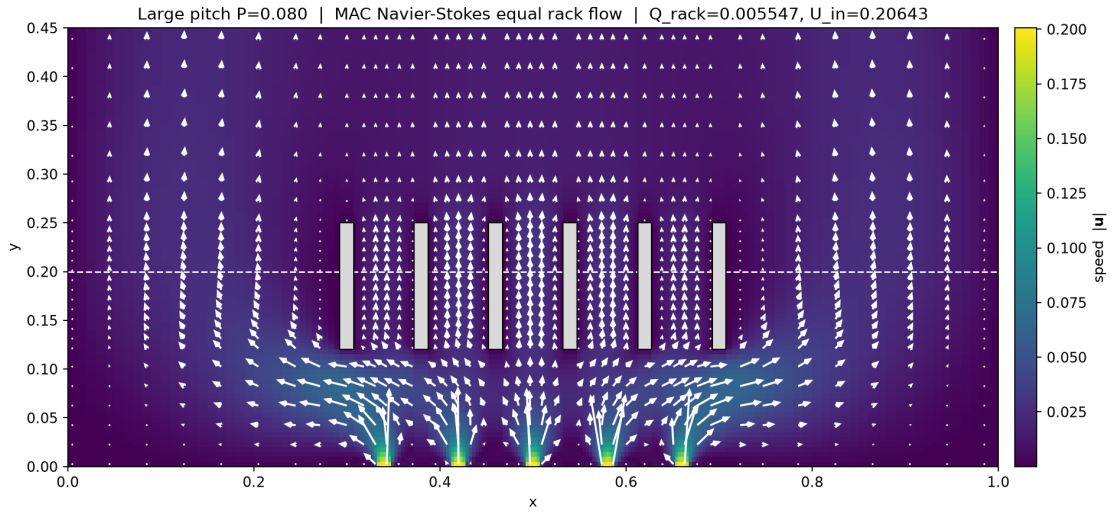


Figure 2: Large-pitch ($P = 0.080$) velocity vector field; colour is speed $|u|$, white arrows the velocity, gray blocks the wafers, dashed line the $y_{cut} = 0.20$ flux line.

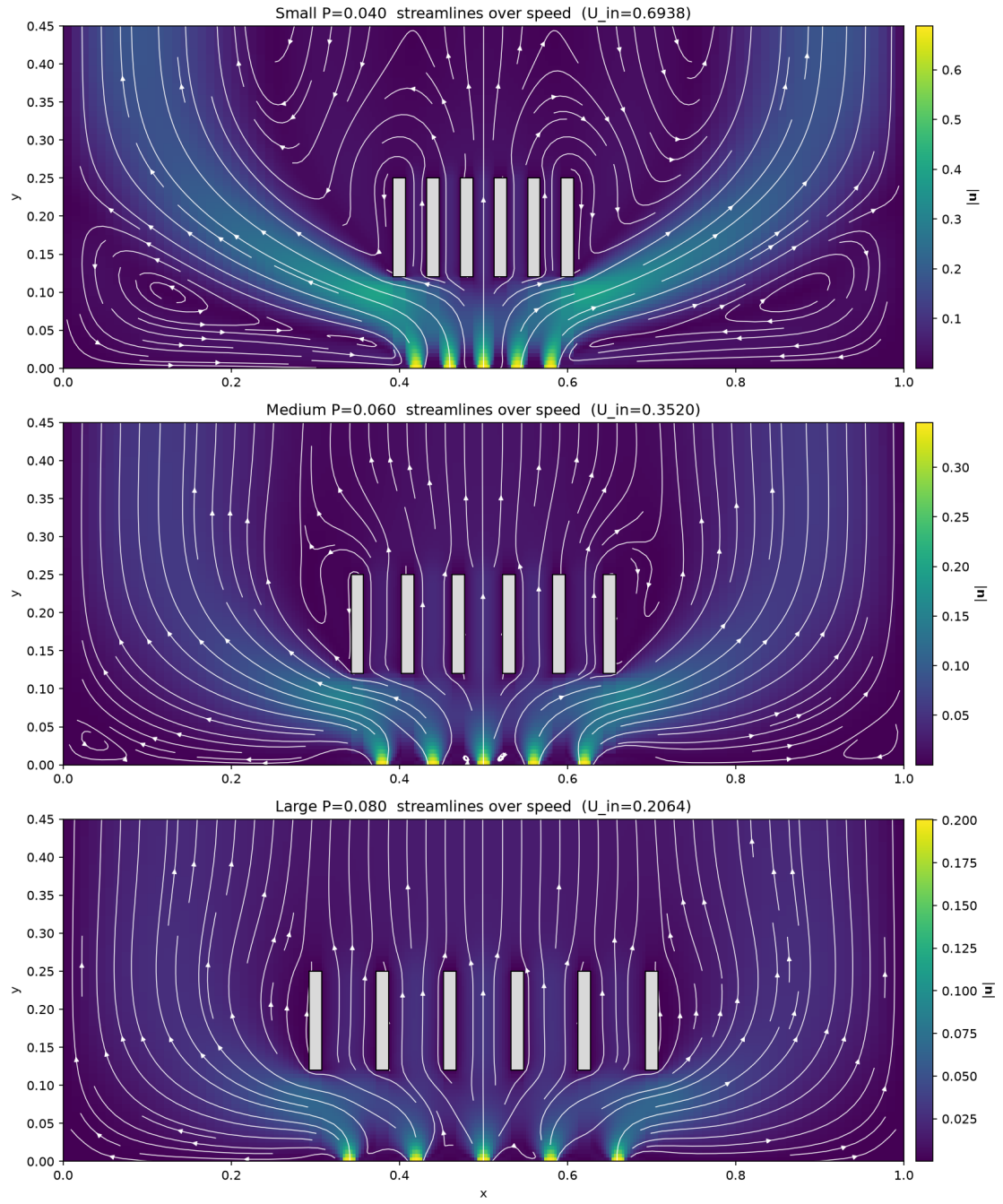


Figure 3: Streamlines over speed for the three pitches (velocity interpolated to a uniform grid for streamline tracing).

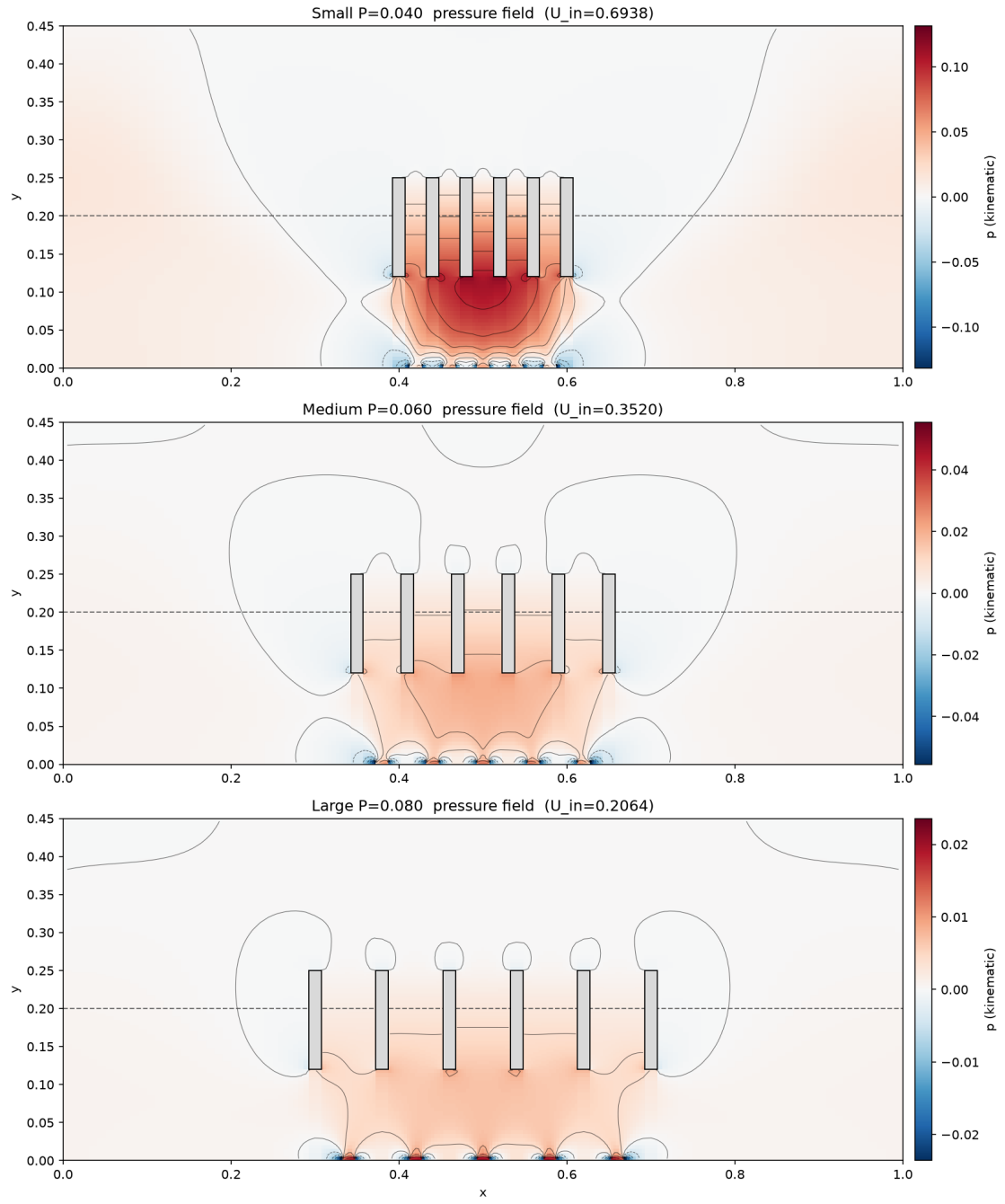


Figure 4: Kinematic pressure field with contours. Pressure drop across the rack grows sharply as the pitch tightens, consistent with the R_h ordering.

8 Equal- Q cut profiles

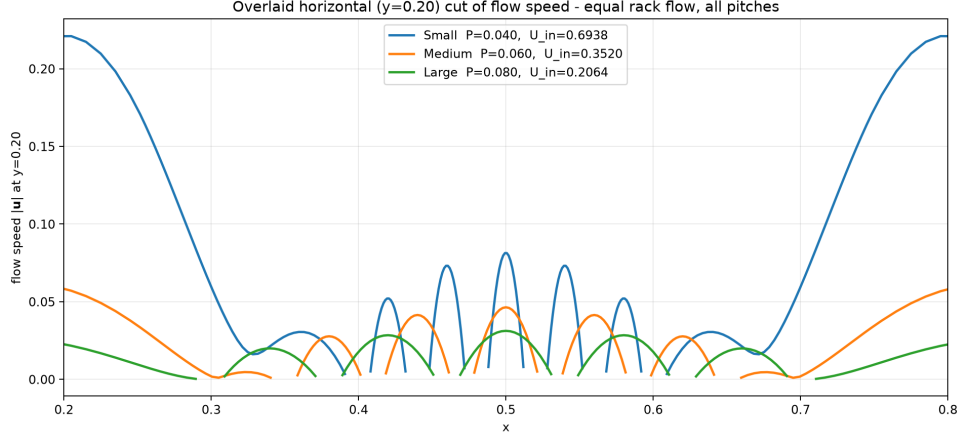


Figure 5: Overlaid $y_{\text{cut}} = 0.20$ cut of flow speed. Tighter pitch reaches a higher peak speed in the clear gaps to carry the common Q_{rack} .

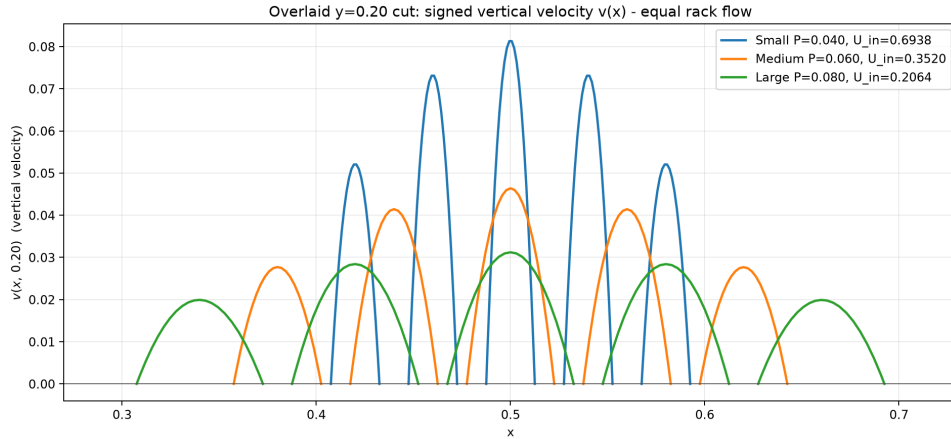


Figure 6: Overlaid $y_{\text{cut}} = 0.20$ cut of signed vertical velocity $v(x)$ over the five interior gaps (no-slip zeros imposed at wafer faces).

9 Interpretation of the comparison

The equal- Q condition equalizes flow through the *five interior gaps*, not the total inlet flow: a substantial and pitch-dependent fraction of the inlet stream bypasses outside the rack span (visible as side flow in the field plots), so only a minority of Q_{in} passes through the interior gaps, and that minority fraction *grows* with pitch. The result should therefore be read as: the inlet speed is tuned per geometry to produce the same interior-gap flow – not as a comparison at equal pump flow or equal total inlet flow. On the corrected (grid-aligned) geometry, the interior gaps carry only 10.7% of the inlet flow for Small, 21.0% for Medium and 35.8% for Large; the remainder bypasses outside the rack span (visible as side flow in the field plots), and the interior-gap share grows with pitch.

10 Caveats and open validation items

The following limitations bound how the quantitative results may be used:

- **Discrete inlet widths (corrected).** The inlet slot edges are now exact x -grid faces, so every discrete slot is 0.0150 m and the total inlet width is 0.0750 m for all pitches. This removed the earlier 10–13% under-sizing and lowered U_{in} for Medium and Large accordingly.
- **Approximate steady state.** See §3: acceptance is a tightened, enforced steady-state gate (not an exact converged state).
- **Pressure-tap sensitivity.** R_h depends on the tap planes (§4). On the corrected Large field, the default planes ($y \approx 0.0975/0.2825$) give $R_h = 0.987$; moving the tap planes by a few grid rows to $y = 0.1175/0.2525$ (just outside the wafer ends) gives 1.070 (+8.4%) and to $y = 0.0775/0.2975$ gives 0.945 (−4.2%). Only the ordering is tap-robust.
- **Grid/timestep convergence (Small, preliminary).** A three-mesh refinement and a timestep halving were run for the Small case (§6): timestep error is negligible ($< 0.02\%$) and grid refinement is monotonic (last successive-grid change -0.64% in R_h). The continuum-limit error is *not* formally estimated from three nonuniform grids, so no “ R_h accurate to $X\%$ ” claim is made. Medium and Large have not been refined; their absolute values carry at least comparable, unquantified grid uncertainty. The large separation between the three R_h makes the ordering robust regardless.
- **Reproducibility, not independent verification.** The rerun harness imports the production Solver; it checks reproducibility and cannot expose an error shared by that implementation.

11 Conclusions

The MAC projection solver is discretely consistent and mass-conserving to roundoff. The equal- Q root solve reliably drives Q_{rack} to the common target, and the corrected code gives hydraulic resistances $R_h = 14.740 > 2.728 > 0.987$ (Small > Medium > Large) *for the present discretization*, essentially unchanged by the inlet correction. With the inlet slots grid-aligned, the corrected inlet speeds are $U_{\text{in}} \approx 0.694/0.352/0.206$ (Small unchanged; Medium and Large lower than the pre-alignment 0.388/0.237, because the earlier slots were 10–13% undersized). These remain reproduced computations, not mesh-independent, physically validated quantities (§10). A preliminary grid and timestep study has been performed for Small only (monotonic refinement, negligible timestep error, continuum error not formally quantified); Medium and Large remain unrefined, so their absolute U_{in} , Δp and R_h are not yet claimed discretization-independent. The absolute speeds also differ from the legacy finite-difference model’s (0.5605/0.3000/0.1833) because the numeric target Q_0 was carried over as a fixed constant rather than redefined self-consistently for the MAC discretization.